

Performance Testing

Date	12 JULY 2026
Team ID	N/A - Individual Submission
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual browser/API checks, FastAPI docs, local smoke testing approach
Type of Testing	Basic load readiness and response validation for demo environment
Target Module / API	/auth, /loans, /analysis/financial-health, /settlement/predict, /settlement/letter
Test Environment	Local Windows development environment
Test Date	12 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Login/register and open dashboard	1-5	60 sec	Successful token generation and dashboard render.
2	Submit financial health analysis repeatedly	5	120 sec	Average response below 2 seconds.
3	Run settlement prediction with varied balances	5	120 sec	Valid settlement, percentage and risk level returned.
4	Generate negotiation letters 1-5	3	90 sec	Letter returned and saved to history.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	Under 1 second in local demo checks	Pass	Logic is lightweight and local.
2	Response Time (Max)	< 5 seconds	Under 2 seconds in local demo checks	Pass	No external AI/API dependency.
3	Throughput (Req/sec)	Supports demo usage	Suitable for single-user evaluation	Pass	Production load testing is future work.
4	Error Rate	< 1%	No functional errors found in normal path	Pass	Invalid input handled by validation
5	CPU Utilization	< 80%	Low in local demo use	Pass	Small FastAPI/React app.
6	Memory Utilization	< 80%	Low in local demo use	Pass	Small FastAPI/React app.